Supplementary Material: When Retrieval Quality Decouples from Faithfulness: A Pre-Registered Audit of RAG Evaluation

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1 Pre-Registration

The matched-context, scaled, metric-fragility, threshold-transfer, noise, cost-aware-baseline, and human-calibration audits were pre-registered on 2026-04-26 before any results were inspected. (The human-calibration design—stratified $n = 100$, two annotators, Cohen's κ—was specified as the optional human-annotation arm of the metric-fragility pre-registration, and was executed at $n = 99$ within two days of the freeze.) The pre-registration fixed: the matched-pair construction (similarity gap ≤ 0.02, CCS gap ≥ 0.20); the paired Wilcoxon test direction (HIGH>LOW); the 10,000-resample bootstrap for the CI on the mean paired difference; the Wilson 95% CI on hallucination rates; the seed list $\{41, 42, 43, 44, 45\}$; the per-seed paired Wilcoxon as the primary significance test

in the scaled audit; the three faithfulness scorers (DeBERTa proxy, `roberta-large-mnli` proxy, RAGAS-style) plus the fixed-context rescore protocol; the τ-sweep range; and the 200-query head-to-head sample size. The four follow-up arms were added after the pre-registered runs as post-hoc robustness checks: answer-span/control diagnostic, stronger-retriever sanity check, Qwen2.5 second-generator replication, and long-form stress test. We disclose which experiments are pre-registered (`audit_log.md` sections under "Pre-Registered") versus follow-ups (`audit_log.md` sections under "Follow-Up") in the file `audit_log.md` shipped at the top level of the artifact.

2 Scaled Audit: Per-Seed Detail

The scaled audit on SQuAD/Mistral runs $n = 2{,}500$ examples per condition for each of five seeds; pooled means appear in the main paper. Table 1 reports per-seed paired Wilcoxon p-values, mean paired differences, and Cohen's d_z for each contrast.

Table 1: Per-seed paired contrasts on SQuAD/Mistral, $n = 500$ pairs per seed. Significance is paired Wilcoxon at $\alpha = 0.05$.

Seed	Contrast	Mean diff	Wilcoxon p	Cohen's d_z
41	baseline−v1	+0.0107	0.083	+0.072
42	baseline−v1	+0.0063	0.234	+0.043
43	baseline−v1	+0.0078	0.174	+0.055
44	baseline−v1	+0.0204	**0.008**	+0.134
45	baseline−v1	+0.0073	0.203	+0.048
41	v2−v1	+0.0132	**0.025**	+0.088
42	v2−v1	+0.0030	0.504	+0.021
43	v2−v1	+0.0059	0.215	+0.042
44	v2−v1	+0.0191	**0.008**	+0.126
45	v2−v1	+0.0125	**0.036**	+0.081

Bootstrap CIs on the pooled mean paired difference for baseline−v1 include zero, and only 1 of 5 seeds reaches significance for that contrast. The v2−v1 contrast reaches significance in 3 of 5 seeds, but the magnitude is small (per-seed mean diff 0.003–0.019). Seed-level uncertainty is treated as a ControlledRAG reporting axis, not a result to average away.

3 Answer-Span/Control Diagnostic Details

The answer-span detector lowercases text, strips punctuation, normalises whitespace, and checks whether any normalised gold-answer alias appears in the retrieved context for the audited RAG setting [Lewis et al., 2020]. It uses no external model. The resulting control rows and model summaries are written under the Fix 12 data and results directories.

Table 2: Matched-context span and outcome summary.

Group	n	span present	faithfulness	hallucination	CCS
all	400	0.220	0.637	0.128	0.404
HIGH-CCS	200	0.170	0.636	0.165	0.670
LOW-CCS	200	0.270	0.639	0.090	0.137
span absent	312	0.000	0.628	0.119	0.408
span present	88	1.000	0.670	0.159	0.390

Table 3: Full span/control model comparison. CIs are bootstrap 95% intervals from the script output.

Predictors	R^2	R^2 CI	AUROC	AUROC CI
answer span	0.019	$[0.001, 0.062]$	0.531	$[0.502, 0.599]$
mean similarity	0.002	$[0.000, 0.023]$	0.586	$[0.495, 0.677]$
CCS	0.003	$[0.000, 0.022]$	0.639	$[0.564, 0.712]$
span + similarity	0.024	$[0.004, 0.071]$	0.590	$[0.518, 0.687]$
span + CCS	0.022	$[0.003, 0.068]$	0.631	$[0.572, 0.716]$
span + similarity + CCS	0.025	$[0.006, 0.075]$	0.643	$[0.589, 0.734]$

The headline reading is reported in the main paper: span presence is the strongest single predictor of continuous faithfulness, CCS is the strongest single predictor of hallucination AUROC, and the joint feature set has the highest score on both. The R^2 values are uniformly small in absolute terms; the relative ranking is the load-bearing signal. Span presence is a weak lexical proxy, not a gold support variable: retrieved text can contain the answer string while still failing to support the generated answer. The mean-score and thresholded hallucination views also inspect different parts of the score distribution.

Table 4: Matched HIGH/LOW CCS faithfulness distribution.

Group	n	mean	p10	p25	p50	p75	p90
HIGH-CCS	200	0.636	0.476	0.553	0.638	0.713	0.795
LOW-CCS	200	0.639	0.512	0.569	0.648	0.715	0.769

Table 5: Matched HIGH/LOW CCS hallucination rates as the score threshold varies.

Group	$\theta = 0.3$	$\theta = 0.4$	$\theta = 0.5$	$\theta = 0.6$	$\theta = 0.7$
HIGH-CCS	0.0%	3.0%	16.5%	38.5%	71.0%
LOW-CCS	1.0%	3.0%	9.0%	38.5%	70.0%

These tables explain why the matched mean contrast can be null while a specific thresholded hallucination contrast differs: the distributions are close in the middle, and the observed HIGH/LOW difference is threshold-sensitive rather than a clean directional CCS effect.

4 Stronger Retriever Sanity Check

The retriever sanity check has two parts. First, the matched contexts from the main-paper audit are re-encoded with MiniLM, BGE-large, and E5-large; this is a fixed-context diagnostic, not a fresh generation run. Second, the completed multi-retriever generation artifact is copied into `data/revision/fix_13/` and summarised. The fixed generations are unchanged across embedders, so the matched HIGH−LOW faithfulness contrast inherits the matched-context null from the main paper and is not re-tested here.

The safe interpretation is retriever sensitivity, not universal robustness. Stronger embeddings change the retrieval surface, sometimes in a direction that flips the SQuAD paradox sign. Controlled-dRAG asks for retriever identity to be reported as a routine audit axis; the supplement here is the operational example.

Table 6: Fixed-context re-encoding of the matched HIGH/LOW CCS pairs. The HIGH>LOW CCS ordering is preserved across all three embedders.

Embedder	pairs	CCS high	CCS low	high−low	sim gap
MiniLM	200	0.670	0.137	0.533	0.006
BGE-large	200	0.739	0.491	0.249	0.032
E5-large	200	0.830	0.702	0.127	0.015

Table 7: Paradox magnitude by embedding model from the completed multi-retriever generation artifact ($n = 30$ per cell, original pilot sampling protocol re-run under different embedders). The SQuAD column shows that the paradox direction depends on retriever; PubMedQA preserves a positive paradox across all four embedders. *These rows are not directly comparable to the $n = 2{,}500$ scaled-audit number in main paper §6 (baseline$-v1 = 0.011$): that table uses the audit-construction sampling and a single MiniLM retriever, whereas this table uses the small-n pilot sampling and varies the embedder. Cross-table comparison should be same-row across embedders within this table, not this table vs. the main-paper scaled-audit table.*

Dataset	embedder	baseline−v1	v2−v1
SQuAD	MiniLM	+0.075	+0.099
SQuAD	BGE-large	+0.032	+0.025
SQuAD	E5-large	−0.011	+0.008
SQuAD	GTE-large	−0.001	−0.006
PubMedQA	MiniLM	+0.024	+0.019
PubMedQA	BGE-large	+0.033	+0.028
PubMedQA	E5-large	+0.053	+0.039
PubMedQA	GTE-large	+0.006	+0.002

5 Second-Generator Metric-Fragility Replication

The second-generator audit samples $n = 100$ fixed contexts per condition from the matched-context construction and generates answers with Qwen2.5 via local Ollama. Those fixed generations are scored with the DeBERTa proxy, the `roberta-large-mnli` proxy, and a local RAGAS-style judge (Mistral via Ollama).

Table 8: Qwen2.5 fixed-generation metric means and raw score contrasts.

Metric	baseline	HCPC-v1	HCPC-v2	baseline−v1	v2−v1
DeBERTa	0.645	0.631	0.644	+0.015	+0.014
roberta-large-MNLI	0.362	0.316	0.368	+0.046	+0.051
RAGAS-style	0.806	0.672	0.814	+0.134	+0.142

Table 9: Metric correlations on the Qwen2.5 fixed-generation rows ($n = 300$).

Metric pair	Pearson r	Spearman ρ
DeBERTa vs. roberta-large-MNLI	0.380	0.429
DeBERTa vs. RAGAS-style	0.280	0.318
roberta-large-MNLI vs. RAGAS-style	0.721	0.668

The scorer-disagreement ordering is preserved under this compact second-generator replication. The claim is limited to this audited Qwen2.5 cell; ControlledRAG explicitly asks for a broader

generator panel before any universal-generator claim is made.

6 Scorer Implementation and Standardized Contrasts

The frozen artifact uses two legacy NLI-style zero-shot label proxies and one context-conditioned RAGAS-style judge. The DeBERTa proxy (`cross-encoder/nli-deberta-v3-base`) and RoBERTa proxy (`roberta-large-mnli`) score each generated answer sentence as the sequence to be classified against the labels entailment/neutral/contradiction with `hypothesis_template="{}"`. The retrieved context is retained in the per-row data and used by the RAGAS-style judge, but it is not consumed by those two legacy zero-shot proxy calls. This disclosure is part of the scorer audit: model identity, input formatting, and prompt format are not implementation trivia.

Because the three scorers are on different native scales, the main paper reports raw contrasts as score-scale-dependent contrasts and adds a standardized Mistral analysis. Table 10 z-scores each scorer over the same 7,500 fixed-generation rows and also reports percentile-rank-normalized contrasts. CIs are paired bootstrap intervals over matched query/seed keys; 2,499 paired keys are available after one duplicated key is collapsed by the pivot.

Table 10: Standardized fixed-generation scorer-fragility contrasts on Mistral rows. Positive values mean the left condition scores higher than HCPC-v1.

Scorer	Contrast	raw diff	z diff (CI)	rank diff (CI)
DeBERTa proxy	baseline−v1	0.010	0.071 [0.032, 0.110]	0.022 [0.010, 0.033]
DeBERTa proxy	v2−v1	0.011	0.073 [0.034, 0.112]	0.022 [0.011, 0.034]
roberta-large-MNLI proxy	baseline−v1	0.032	0.231 [0.191, 0.271]	0.062 [0.051, 0.074]
roberta-large-MNLI proxy	v2−v1	0.032	0.236 [0.199, 0.275]	0.063 [0.052, 0.075]
RAGAS-style judge	baseline−v1	0.139	0.337 [0.298, 0.375]	0.085 [0.075, 0.095]
RAGAS-style judge	v2−v1	0.138	0.332 [0.294, 0.370]	0.083 [0.073, 0.092]

The standardized table preserves the scorer-fragility conclusion without treating raw scores on different scales as directly comparable causal effects.

7 Long-Form/Synthesis Stress Test

The long-form stress test packages an existing QASPER/MS-MARCO run with 40 distinct long-form questions. It is included only as an exploratory scope probe, not as broad generalization evidence for the short-answer audit.

Table 11: Long-form stress-test condition means.

Dataset	condition	span faith	claim faith	unsupported	long halluc.
MS-MARCO	baseline	0.645	0.646	0.224	0.400
MS-MARCO	HCPC-v1	0.678	0.679	0.075	0.050
MS-MARCO	HCPC-v2	0.674	0.674	0.158	0.250
QASPER	baseline	0.649	0.649	0.158	0.300
QASPER	HCPC-v1	0.647	0.647	0.117	0.200
QASPER	HCPC-v2	0.648	0.651	0.129	0.250

The short-answer paradox does not generalise cleanly here: MS-MARCO has a negative span-faithfulness paradox drop (−0.033), and QASPER is near zero (+0.002). The long-form result is reported as a scope limit rather than as broad long-form evidence for ControlledRAG.

8 Self-RAG: Harness-Mismatched Appendix Baseline

We include Self-RAG [Asai et al., 2024] only as an appendix baseline because the available Self-RAG harness mismatches the audit's fixed-context protocol along three dimensions: (i) Self-RAG's generator emits its own retrieval and reflection tokens, so the "fixed retrieved context" assumption used by the matched-context and metric-fragility audits is not directly satisfied; (ii) Self-RAG's recommended generator is a fine-tuned Llama-2-7B variant that does not coincide with the audited Mistral-7B / Qwen2.5-7B family and that was OOM-killed on the available T4 instance for the PubMedQA cell, so the partial SQuAD numbers we have are not directly comparable to the cost-aware head-to-head; and (iii) the recommended Self-RAG threshold differs from the threshold-transfer sweep used elsewhere. We report this disclosure here so that future ControlledRAG-style audits can budget for a matched Self-RAG harness; we explicitly do not generalize from these mismatched runs to Self-RAG overall.

9 Coherence vs. Generic Retrieval Noise

The noise audit uses the matched-context construction as a backbone and replaces 1/3, 2/3, or 3/3 of the retrieved chunks with one of two noise types: random off-topic passages and coherence-preserving uninformative passages drawn from the same topic distribution that do not contain the answer.

Table 12: Faithfulness and similarity slopes per noise rate.

Condition	faith slope (per noise rate)	sim slope (per noise rate)
random noise	−0.069	−0.481
coherent uninformative noise	−0.043	−0.113
HCPC-v1 refinement (reference)	—	—

The two noise types produce distinguishable faithfulness slopes at matched substitution rate; we do not claim the gap is large in absolute terms. The right reading is that coherence and similarity are correlated but not identical.

10 Threshold Transfer: Full Sweep

The main paper reports a per-tune-dataset summary (diagonal recovery, off-diagonal mean, gap) at each tune dataset's selected τ^*. The full 5×5 per-cell transfer matrix at each tune dataset's τ^* is reproduced below; the per-τ sweep over $\{0.3, 0.4, 0.5, 0.6, 0.7\}$ for each dataset is in the artifact's `results/revision/fix_04/` directory. Recovery is the dimensionless normalized contrast

$$R_\tau(d) = \frac{\text{faith}_{\text{gate},\tau}(d) - \text{faith}_{\text{HCPC-}v1}(d)}{\text{faith}_{\text{baseline}}(d) - \text{faith}_{\text{HCPC-}v1}(d)}.$$

It is useful for asking whether a tuned threshold transports, but it is not a direct utility measure: when the denominator is small or negative, the sign can be counterintuitive and values can exceed 1.

Per-τ recovery curves and confidence diagnostics for the τ-tuning step are included as machine-readable CSVs in the same results directory. The transfer matrix itself does not include bootstrap CIs, so we use it as a descriptive transport diagnostic.

Table 13: Full 5×5 threshold-transfer matrix. Each row gives the normalized recovery ratio of the τ tuned on the row dataset, evaluated on each of the five column datasets. Positive values are not always "better" when the baseline−v1 denominator is negative. Rows can be byte-identical across the eval columns when two tune datasets pick the same τ^*: tune-SQuAD and tune-HotpotQA both pick $\tau^* = 0.3$, and tune-PubMedQA and tune-TriviaQA both pick $\tau^* = 0.4$, so the off-diagonal columns coincide. This is a faithful consequence of the experiment (off-diagonal recovery only depends on the eval dataset and the selected τ, not on which tune dataset selected it), not a pipeline bug.

τ tuned on	τ^*	SQuAD	PubMedQA	HotpotQA	NaturalQs	TriviaQA
PubMedQA	0.4	+0.489	+1.452	−0.291	+1.140	+0.464
NaturalQs	0.5	+0.327	+0.664	−0.471	+1.401	+0.323
SQuAD	0.3	+0.823	−1.714	−0.112	+0.865	+0.398
TriviaQA	0.4	+0.489	+1.452	−0.291	+1.140	+0.464
HotpotQA	0.3	+0.823	−1.714	−0.112	+0.865	+0.398

11 Human Calibration Protocol

The human-calibration panel asks two raters (one expert, one general-domain) to label $n = 99$ matched-cell items with a binary faithful/hallucinated decision and a 0–1 confidence score. The panel is designed for calibration of automated scorers against a reference human label set on the audited cell, not for a settled scorer ranking.

Protocol. We use rater 1's labels as the reference human label set; rater 2 independently re-labeled the same 99 items as a reliability check. Cohen's $\kappa = 0.774$ between the two label sets is reported as inter-rater reliability rather than as a disagreement-resolution process. We do not collapse the two raters into a third "resolved" label by majority or by discussion; with two raters either procedure would be ill-defined. Spearman correlations against the rater-1 label set are $\rho_{\text{DeBERTa}} = 0.140$, $\rho_{\text{second NLI}} = 0.380$, $\rho_{\text{RAGAS-style}} = 0.441$. Bootstrap 95% CIs on the three correlations overlap, so the metric ranking is suggestive rather than settled.

12 Source Trace and Artifact Manifest

Every numerical claim in the main paper traces to a result file under `results/revision/` with a corresponding pre-registered script under `experiments/`. Table 14 summarises the trace; the full mapping is shipped as `SOURCE_TRACE.md` in the artifact and reproduced here for the reviewer's convenience.

Reproducibility. An anonymized artifact package containing code, per-query CSVs, matched context pairs, source tables, scripts, and documentation is provided for review. Public package, benchmark, and permanent archive links are omitted during double-blind review and will be released upon acceptance.

Table 14: Audit trace from claim to source file.

Audit	Source results files
Matched-context (§3 backbone)	`results/revision/fix_01/`
Scaled audit (§2)	`results/revision/fix_02/`
Metric fragility (Mistral)	`results/revision/fix_03/`, `standardized_scorer_fragility.csv`
Threshold transfer (§10)	`results/revision/fix_04/`
Noise (§9)	`results/revision/fix_05/`
Cost-aware head-to-head	`results/revision/fix_06/`, `h2h_summary_with_ci.csv`, `fix_11/`
Span/control (§3)	`results/revision/fix_12/`
Stronger retrievers (§4)	`results/revision/fix_13/`, `results/multi_retriever/`
Second generator (§5)	`results/revision/fix_14/`
Long-form stress (§7)	`results/revision/fix_15/`
Human calibration (§11)	`results/human_*/`

13 Resume and Compute Notes

The local scripts are resumable through their CSV outputs. The only script likely to benefit from a free-tier GPU notebook is the second-generator replication if local Ollama is slow: run the same command on a free GPU-backed notebook after pulling Qwen2.5 and Mistral into Ollama, then download `data/revision/fix_14/` and `results/revision/fix_14/`. None of the audit experiments require paid APIs; all scorer rescoring is local DeBERTa, `roberta-large-mnli`, and a Mistral-via-Ollama RAGAS-style judge.

References

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